

Weld Engineering Report

Professional estimate of weld geometry, weld metal, filler metal consumption, and process-based arc-on time for planning and review.

Joint: Pipe Butt Weld**Groove:** Double V**Process:** GTAW + SMAW**Classification:** AWS A5.18 + AWS A5.1 ER70S-2 + E7018**Family:** Carbon Steel**Generated:** 2026-06-30 21:09**ESTIMATE READY**

Estimated filler metal consumption is 4.461 kg with 3.054 h of arc-on time.

Effective Deposition Rate: 1.46 kg/h

Filler per Meter: 1.096 kg/m

Arc-On per Meter: 45.01 min/m

Executive Summary

Primary estimate outputs prepared for engineering review and shop planning.

Weld Area

89.57 mm²

Weld Length

4070.25 mm

Weld Metal Volume

364.569 cm³

Weld Metal Weight

2.862 kg

Filler Metal Consumption

4.461 kg

Estimated Arc-On Time

3.054 h

Effective Deposition Efficiency

70.6%

Effective Deposition Rate

1.46 kg/h

Planning Indicators

Normalized performance indicators for comparing joints, labor load, and consumable demand.

Filler per Meter

1.096 kg/m

Weld Metal per Meter

0.703 kg/m

Arc-On per Meter

45.01 min/m

Filler per Joint

1.115 kg/joint

Arc-On per Joint

45.80 min/joint

Efficiency Loss Basis

1.193 kg

Waste Allowance Basis

0.406 kg

Consumption Multiplier

1.559 x

Process Breakdown

Split estimate showing deposited weld metal, filler demand, and arc-on time by process segment.

Process	Area Share	Weld Metal (kg)	Filler (kg)	Arc-On Time (h)	Rate (kg/h)	Efficiency
GTAW	25.0%	0.715	0.828	1.036	0.80	95.0%
SMAW	75.0%	2.146	3.632	2.018	1.80	65.0%

Engineering Basis

Input selections and governing geometry used to calculate the estimate.

Setup and Assumptions

Parameter	Value
Process	GTAW + SMAW
Rate Basis	Estimated
Joint	Pipe Butt Weld
Geometry	Equal
Groove	Double V
Classification	AWS A5.18 + AWS A5.1 ER70S-2 + E7018
Filler Metal Family	Carbon Steel
Density	7.85 g/cm3
Waste Allowance	10%
Quantity	4

Joint Geometry

Parameter	Value
Pipe OD	323.9 mm
Thickness	16 mm
Root Gap	3 mm
Root Face per Side	2 mm
Bevel Angle	30 deg

Process Parameters

Parameter	Value
GTAW Transition Depth	4 mm
GTAW Wire Diameter	2.4 mm
SMAW Electrode Diameter	4.0 mm

Calculation Method

Formula basis used in the application for weld volume, weld metal, filler consumption, and arc-on time.

$$\text{Volume (cm3)} = \text{Area (mm2)} \times \text{Length (mm)} / 1000$$

$$\text{Weld Metal (kg)} = \text{Volume (cm3)} \times \text{Density (g/cm3)} / 1000$$

$$\text{Filler Consumption (kg)} = \text{Weld Metal} / \text{Deposition Efficiency} \times (1 + \text{Waste} / 100)$$

$$\text{Arc-On Time (h)} = \text{Filler Consumption} / \text{Deposition Rate}$$

Engineering Notes

Practical interpretation notes for planning, estimating, and report handoff.

- Arc-on time covers active welding time only. Fit-up, tacking, interpass cleaning, repositioning, and inspection time are excluded.
- Filler metal consumption includes deposition efficiency loss and the entered waste allowance. It should be treated as planning consumption, not exact issued weight.
- Combined GTAW + SMAW output distributes weld metal and time by calculated process share using the entered transition depth.
- This report is intended for estimating and engineering planning. Approved project documentation, client specifications, and production controls must always take precedence.